

Foundational IT Skills

A practical engineering-student handbook for computer hardware, Linux file management, professional office tools, data analysis, cloud collaboration, responsible generative AI and digital design.

POLY PMNA

Course code	1008
Programme	Diploma in Engineering and Technology
Semester	I
Type	Lab
Teaching scheme	0:0:2:0
Contact hours	30
Self-learning	15
Credits	1
CIE / ESE	60 / 40

Curriculum identity and source control

Official source used

The curriculum authority for this handbook is the supplied five-page SITTTTR Kerala **Diploma Curriculum Revision 2026** syllabus PDF for **Course 1008 - Foundational IT Skills**. No Revision 2021 course content has been merged.

Programme / department

Diploma in Engineering and Technology. A specific branch or department is not specified in the supplied PDF.

Course category

Not specified in the supplied official syllabus PDF.

Prerequisite

NIL.

Course description

Lab course bridging basic computer usage and professional engineering standards through office automation, cloud collaboration, data visualization and Generative AI tools.

Verified inconsistency / ambiguity

The overall assessment is consistently shown as **CIE 60 + ESE 40 = 100**. However, the page-4 Assessment Methodology table is internally ambiguous: it prints CE exam mark 50, CA1 40 converted to 15, CA2 40 converted to 15, OEE 50 converted to 10 and a separate “Marks” row showing Attendance 5, CE 30, OEE 10 and ESE 40. Those displayed components do not form one unambiguous arithmetic breakdown of the 60 CIE marks.

Spreadsheet-function inconsistency: the Detailed Syllabus prints “AVG”, while the self-learning activity prints “AVERAGE”. Major spreadsheet tools use AVERAGE as the executable function, so the handbook preserves the source wording in declarations but teaches =AVERAGE (range) .

COUNT technical mismatch: the official cricket activity asks students to use COUNT on the text-based Player Name range A2:A21. COUNT normally ignores text, so it may return 0. The handbook retains the activity and explicitly teaches COUNTA for verifying non-empty player names.

Display decision: this handbook uses the official overall totals 60/40 for navigation and preserves the component rows exactly as printed instead of inventing a corrected distribution. Faculty instructions and the current institutional assessment circular must control actual mark entry.

No hour conflict found: 12 experiments = 24 hours, two Series Tests = 4 hours and the Open-ended Experiment = 2 hours, giving the official 30 contact hours. Self-learning activities total 15 hours.

COURSE DASHBOARD

Scheme, purpose and outcomes at one glance

30

contact hours

15

self-learning hours

1

credit

12 + 1

prescribed experiments + open-ended work

Official teaching and assessment scheme									
L	T	P	S/R	Self-learning / week	Total hours / semester	Credits	Practical CIE	Practical ESE	Total
0	0	2	0	1	30	1	60	40	100

Course objectives - official wording

- 1. Familiarize the students with basic functions and features of Computers, Operating systems and Internet applications.
- 2. Enable the students to prepare documents, spreadsheets and presentations.
- 3. Equip students with data visualization and automation skills using Pivot tables, slicers, graphs and charts.
- 4. Empower students to use generative AI tools for ethical content creation and guided technical learning.

How to use this handbook

- 1. Read the concept explanation before opening software.
- 2. Study the diagram, Malayalam support and quality warning.
- 3. Follow the experiment procedure and record evidence.
- 4. Compare your output with the observation and result criteria.
- 5. Attempt the expected questions without opening answers.
- 6. Use the Rule Bank and Quick Revision before tests.

Official Course Outcomes		
CO	Official outcome	Bloom level
CO1	Identify the basic functions and features of Computers, Operating Systems, and Internet applications.	Understand
CO2	Use Stand-alone and Cloud-based office tools to prepare documents, spreadsheets, and presentations.	Apply
CO3	Apply spreadsheet functions and Generative AI tools to construct interactive dashboards, automated reports, and professional visual presentations for interpreting complex datasets.	Apply
CO4	Apply generative AI, digital design platforms for guided learning and visual communication.	Apply

COURSE ARTICULATION MATRIX

Official CO-PO mapping

Correlation: 1 Low, 2 Medium, 3 High, - No correlation											
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	3	3	-	-	-	-
CO3	3	3	-	3	-	-	3	-	-	-	-
CO4	3	3	3	3	-	-	3	-	-	-	-
Total	12	6	3	6	-	3	9	-	-	-	-
Correlation level	3	3	3	3	-	3	3	-	-	-	-

CURRICULUM COVERAGE MAP

Every official area mapped to a handbook section

Coverage matrix verified against the supplied PDF					
Module	Official topics	Hours	CO	Cognitive level	Handbook section
I	Hardware & Software Exploration; Linux GUI file/folder management; advanced web search, email etiquette and LMS/Google Classroom	6	CO1	Remember / Understand	Module I
II	Professional documentation; spreadsheet formulas and references; five-slide presentation; cloud collaboration/version history; Series Test 1	10	CO2	Apply	Module Ii
III	PivotTables and slicers; spreadsheet charts; AI slide generation and verified AI-assisted research	6	CO3	Apply	Module Iii
IV	GPT-guided learning; C-R-T-O prompt framework; graphic design with AI tools; Series Test 2	6	CO4	Apply	Module Iv
Open-ended	Student-designed experiment integrating one or more course concepts	2	CO1 (as printed)	Apply	Open Ended

Learning roadmap

Identify hardware and software

Create professional office outputs

Use AI and design responsibly

Organise files and communicate

Analyse and visualize data

Computer Foundations, Linux Files and Connectivity

Build a reliable base: identify the machine, organise files correctly, search the web precisely and communicate safely.

Official module title: Not specified in the supplied syllabus PDF; this is a descriptive handbook navigation title.

6 instructional hours

CO1

Remember / Understand

Malayalam support: Hardware physical parts ആണ്; Software instructions/programs ആണ്. RAM working memory ആണ്, HDD/SSD permanent storage ആണ്. File structure clear ആക്കി maintain ചെയ്താൽ search, backup, version control എന്നിവ reliable ആകും.

Concept handbook

Computer architecture and the data path

Input devices create data; CPU control unit and ALU process instructions; RAM holds active work; storage preserves files; output devices communicate results. Firmware starts hardware and the operating system manages resources.

CPU, RAM, HDD and SSD

Processor model or clock does not alone determine performance. RAM capacity affects active workload. HDD offers magnetic storage with moving parts; SSD offers flash storage and usually lower latency.

Ports and peripherals

USB carries data and sometimes power; HDMI/DisplayPort carry digital audio/video; VGA is analog video; RJ45 Ethernet supports wired networking; audio jacks, webcams and external drives are peripherals.

System software versus application software

Operating systems, device drivers and utilities are system software. Word processors, spreadsheets, browsers, presentation tools and CAD packages are application software.

Linux GUI file operations

Create, rename, move, copy, search and delete using a file manager. Use a predictable directory tree, clear filenames, versions and backups.

Advanced search and source verification

Combine exact phrases, site:, filetype: and minus operators. Verify authority, date, scope, file type and consistency with another reliable source.

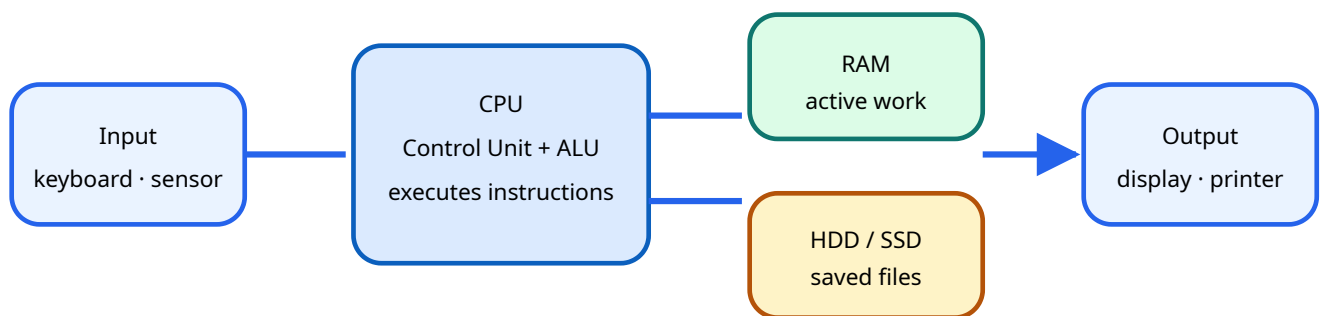
Email fields and netiquette

To is the primary action recipient; CC keeps stakeholders informed; BCC protects recipient privacy. Use a specific subject, concise body, signature and attachment check.

LMS and connectivity

Enroll with the correct identity, read announcements, submit the right file format and confirm submission status. Protect credentials and use secure networks.

Computer system data path



Observe the difference between temporary working memory and persistent storage.

Practical / Activity / Experiment Centre

Experiment 1 Hardware & Software Exploration

2 instructional hours · CO1 · Remember

Official practical outcome: Identify the CPU, RAM, HDD/SSD and peripheral ports; identify application software and system software on a personal computer.

Aim

Identify major computer hardware, peripheral ports, system software and application software without damaging the equipment.

Apparatus / software

Laboratory PC, hardware demo kit, ESD wrist strap when available, port-identification sheet, operating-system system-information utility.

Underlying theory

A computer system combines hardware, firmware and software. CPU executes instructions; RAM is volatile working memory; HDD/SSD provide non-volatile storage. Ports provide physical interfaces, while the operating

Safety, privacy and quality

Shut down and isolate power before opening a cabinet. Do not touch exposed contacts. Use ESD precautions and faculty supervision for disassembled components.

system, drivers and utilities manage hardware resources.

Procedure

1. Record PC make/model and operating system.
2. Open the system-information tool and note processor, installed RAM and storage.
3. Identify motherboard, CPU location, RAM module, HDD/SSD and SMPS from the demo kit.
4. Identify USB-A, USB-C, HDMI, VGA/DisplayPort, audio, Ethernet and power ports.
5. List at least five system software items and five application software items.
6. Classify each item and obtain faculty verification.

Observation / evidence

Create a table with columns: Item/Port, category, visual identifier, function, evidence or specification.

Result

Major hardware, ports, system software and application software are identified and correctly classified.

Possible errors and troubleshooting

- Confusing RAM capacity with storage capacity.
- Calling every program “software” without classifying it.
- Handling components while the system is powered.

Viva questions with answers

Why is RAM called volatile memory?

Its contents are lost when power is removed.

HDD and SSD differ mainly in what way?

HDD uses moving magnetic parts; SSD uses semiconductor flash and normally has lower access time.

What is a device driver?

System software that lets the operating system communicate with specific hardware.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Experiment 2

File and Folder Management in a Linux GUI

2 instructional hours · CO1 · Understand

Official practical outcome: Create folders and files in Linux using a Graphical User Interface (GUI).

Aim

Create, rename, move, copy and delete files and folders using a Linux desktop file manager.

Apparatus / software

Ubuntu LTS or another institution-approved Linux GUI, file manager, text editor.

Underlying theory

A directory tree organizes data hierarchically. Absolute paths begin at the root; relative paths are interpreted from the current location. Copy produces another item, while move changes location or name.

Safety, privacy and quality

Work only inside the assigned student folder. Do not delete system folders or another student's files. Confirm the target before permanent deletion.

Procedure

1. Create a semester folder using a clear name.
2. Inside it, create one folder per subject code and module subfolders.
3. Create the prescribed Backpack folder with Food and Tools subfolders.
4. Inside Tools create flashlight.txt and map.txt.
5. Move map.txt to Food; copy flashlight.txt as spare_light.txt.
6. Delete Food and its contents, then list the remaining Backpack items.
7. Capture a screenshot or record the final tree.

Observation / evidence

Record each operation, source path, destination path and final status. The final Backpack should contain Tools with flashlight.txt and spare_light.txt.

Result

The required Linux GUI directory tree is created and verified.

Possible errors and troubleshooting

- Using ambiguous filenames such as final2-new.

- Moving when a copy was required.
- Deleting the parent folder instead of the selected subfolder.

Viva questions with answers

What is the difference between copy and move?

Copy creates another instance; move relocates the original.

What is a file extension?

A suffix that commonly indicates file type, such as .txt or .pdf.

Why use subject codes as folder names?

They produce a consistent, searchable structure linked to the curriculum.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Experiment 3

Web & Connectivity

2 instructional hours · CO1 · Understand

Official practical outcome: Practise advanced search operators, email etiquette and Google Classroom.

Aim

Find reliable technical resources, use an LMS and send a privacy-conscious formal email.

Apparatus / software

Web browser, internet connection, email account, Google Classroom or approved LMS.

Underlying theory

Search operators narrow results: quotes request an exact phrase, site: limits a domain and filetype: limits a file format. To/CC/BCC control message delivery and recipient visibility. Netiquette requires a clear subject, respectful language, correct attachments and protection of personal addresses.

Safety, privacy and quality

Do not disclose passwords or OTPs. Avoid suspicious links and executable downloads. Use BCC when recipients must not see one another's email addresses.

Procedure

1. Search using filetype:pdf "Computer Engineering" site:edu and record the first suitable result URL and file size.
2. Check source authority, publication date and file type.
3. Enroll in the Foundational IT Skills LMS and customize the profile as permitted.
4. Draft the official workshop schedule-change email.
5. Place the staff coordinator in To, the HoD in CC and student participants in BCC.
6. Proofread, confirm attachment and send only after faculty approval.

Observation / evidence

Record search string, URL, file size, source authority, LMS completion and email field allocation.

Result

A verified PDF resource is located, LMS participation is completed and a formal privacy-conscious email is prepared.

Possible errors and troubleshooting

- Putting all students in To or CC.
- Using vague subjects such as "Important".
- Trusting the first result without verifying source and date.

Viva questions with answers

Why is BCC used?

It hides recipients' addresses from one another.

What does site:edu do?

It restricts search results to the specified domain pattern.

What should a formal subject line contain?

The action/topic and enough context, for example "Workshop schedule changed - 22 July".

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Expected / Practice Questions

Very Short Answer

Differentiate RAM and storage.

Answer / model points

RAM is volatile working memory used by active programs; storage such as SSD/HDD is non-volatile and preserves files after power-off.

Short Answer

Why should student email addresses be placed in BCC in the official scenario?

Answer / model points

BCC prevents recipients from seeing one another's private email addresses while still delivering the same announcement.

Application

Write a search query for PDF resources about Computer Engineering on educational domains.

Answer / model points

filetype:pdf "Computer Engineering" site:edu

Troubleshooting

A copied file cannot be found. What should be checked?

Answer / model points

Check the destination path, current folder, hidden-file setting, filename/extension and whether Move was accidentally used instead of Copy.

Module I summary

Build a reliable base: identify the machine, organise files correctly, search the web precisely and communicate safely. Complete every experiment with evidence, result, troubleshooting notes and faculty verification; memorising menu paths alone is insufficient.

Professional Office Tools and Cloud Collaboration

Convert raw information into structured documents, correct spreadsheets, clear presentations and recoverable collaborative work.

Official module title: Not specified in the supplied syllabus PDF; this is a descriptive handbook navigation title.

10 instructional hours

CO2

Apply

Malayalam support: Manual formatting ആവർത്തിക്കാതെ Styles, formulas, templates, Slide Master എന്നിവ ഉപയോഗിക്കുക. Cloud file-ൽ തെറ്റായ edit വന്നാൽ Version history വഴി editor-നെയും പഴയ version-നെയും കണ്ടെത്താം.

Concept handbook

Document styles and table of contents

Use Heading 1/2 and body styles; modify the style definition once instead of formatting each heading. Insert captions, page numbers and an automatic TOC, then update fields before PDF export.

Professional biodata layout

Use a two-column borderless table for aligned labels and values. Maintain whitespace, concise evidence-based skills and only necessary personal details.

Spreadsheet data discipline

One header row, one record per row, one variable per column, consistent data types and no decorative blank rows inside the dataset.

Cell references

Relative A1 changes when copied; absolute \$A\$1 is fixed; mixed \$A1 or A\$1 locks one dimension. Use references rather than retyping constants.

Core functions

SUM totals numbers; AVERAGE gives arithmetic mean; COUNT counts numeric cells; COUNTA counts non-empty cells; COUNTIF counts matching records; IF returns one of two outputs.

Presentation planning

Define audience, purpose and call to action before designing. Use one message per slide and evidence rather than decoration.

Slide Master, transitions and animations

Slide Master controls repeated elements. Use transitions between slides and animations within slides only when they clarify sequence or emphasis.

Cloud permissions and version history

Choose Viewer, Commenter or Editor. Use comments/suggestions for review and version history to identify or restore edits.

Optional Overleaf/LaTeX

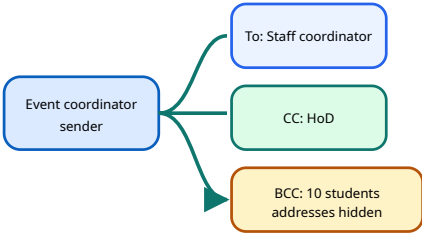
LaTeX source is compiled into PDF. A minimal document includes document class, begin/end document, title, list and math syntax.

Cell-reference behaviour

Row	B: Runs	F: Cut-off	G: Status
2	62	50	=IF(B2>=\$F\$2,...)
3	41	fixed	=IF(B3>=\$F\$2,...)
4	33	fixed	=IF(B4>=\$F\$2,...)

B2 changes when filled down; \$F\$2 remains fixed.

Email recipient logic



BCC is the privacy control in the official announcement scenario.

Practical / Activity / Experiment Centre

Experiment 4

Professional Documentation
2 instructional hours · CO2 · Apply

Official practical outcome: Create a formal report with table of contents and styles.

Aim

Prepare a professional one-page assignment cover sheet and an internship biodata using styles, tables and PDF export.

Apparatus / software

Microsoft Word, LibreOffice Writer or approved equivalent; PDF viewer.

Underlying theory

Styles separate document structure from appearance. Heading styles enable consistent formatting and automatic tables of contents. Tables can align data accurately; borders can be removed while retaining layout.

Safety, privacy and quality

Use only necessary personal information. Do not expose Aadhaar numbers, passwords or sensitive identifiers in sample biodata.

Procedure

1. Set Normal margins: 1 inch or 2.54 cm.
2. Use a professional sans-serif body font.
3. Create Heading 1 and Heading 2 structure; modify Heading 1 to bold and underlined as prescribed.
4. Insert a two-column borderless table for personal details.
5. Add a clearly marked professional-photo placeholder.
6. Insert or update the table of contents when applicable.
7. Inspect page breaks and export to PDF.

Observation / evidence

Check margins, style usage, alignment, table-border state, image placeholder, TOC update and PDF output.

Result

A structured professional document is created and exported as PDF.

Possible errors and troubleshooting

- Formatting every heading manually.
- Typing spaces to align colons.
- Failing to update the TOC before export.

Viva questions with answers

Why are styles better than manual formatting?

They maintain consistency and enable automatic document features.

What generates a TOC automatically?

Properly applied heading styles.

Why use a borderless table for biodata?

It aligns labels and values without visible grid lines.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Experiment 5

Spreadsheet Basics

2 instructional hours · CO2 · Apply

Official practical outcome: Use formulas SUM, AVG/AVERAGE, COUNT, COUNTIF and IF, with cell referencing.

Aim

Create and verify spreadsheet formulas for pass counting, cricket statistics and selection status.

Apparatus / software

Microsoft Excel, LibreOffice Calc, Google Sheets or approved equivalent.

Underlying theory

A formula begins with = and uses cell references. Relative references change when copied; absolute references such as \$F\$1 remain fixed. COUNT counts numeric cells, whereas COUNTA counts non-empty cells.

Safety, privacy and quality

Do not overwrite source data. Keep an unmodified copy and avoid enabling untrusted macros.

Procedure

1. Enter a clean table with one header row.
2. Use =SUM(B2:B21) and =AVERAGE(B2:B21) for runs.
3. Use =COUNTIF(C2:C21,"Batsman") for category count.
4. Use =IF(B2>=50,"Selected","Waitlist") and fill down.
5. Use COUNT on a numeric participant field; explain why COUNT on text names may return zero and use COUNTA when counting names.
6. Check formulas with sample manual calculations and save the file.

Observation / evidence

Record formulas, returned values, copied ranges and manual verification.

Result

Spreadsheet functions and cell references are applied and verified.

Possible errors and troubleshooting

- Using COUNT on a text-only Player Name column and expecting the participant total.
- Omitting quotation marks around text criteria.
- Including the header cell in a numeric range.

Viva questions with answers

COUNT and COUNTA differ how?

COUNT counts numeric cells; COUNTA counts non-empty cells.

What does \$A\$1 mean?

Both column and row are absolute when copied.

What happens when an IF condition is false?

The value_if_false argument is returned.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Experiment 6

Presentation Design

2 instructional hours · CO2 · Apply

Official practical outcome: Design a five-slide pitch deck with animations and transitions.

Aim

Build a clear five-slide pitch deck and use Slide Master to repeat name and roll number.

Apparatus / software

PowerPoint, LibreOffice Impress, Google Slides or approved equivalent.

Underlying theory

A presentation should have one main message per slide, visual hierarchy, consistent alignment and controlled motion. Slide Master stores repeated layout elements across the deck.

Safety, privacy and quality

Use licensed or original media. Avoid inserting confidential screenshots or personally sensitive data.

Procedure

1. Define audience and objective.
2. Create five slides: title, problem, proposed solution, evidence/benefits and call to action.
3. Edit Slide Master so name and roll number appear automatically on every relevant slide.
4. Use readable fonts and high contrast.
5. Apply one restrained transition and only meaningful animations.
6. Run slide show, check timings and export a PDF copy.

Observation / evidence

Record slide count, master elements, font size, contrast, transition and animation purpose.

Result

A consistent five-slide pitch deck is created using Slide Master.

Possible errors and troubleshooting

- Using different styles on every slide.
- Animating every object.
- Reading paragraphs directly from slides.

Viva questions with answers

What is Slide Master?

The master layout controlling repeated design and placeholders.

Why limit animation?

Excess motion distracts from the message and may reduce accessibility.

What is visual hierarchy?

Arrangement that makes the most important information noticed first.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Experiment 7

Cloud Collaboration

2 instructional hours · CO2 · Apply

Official practical outcome: Use real-time editing in Google Docs/Sheets with version history.

Aim

Collaborate on a cloud document and recover an accidentally deleted table using version history.

Apparatus / software

Google Docs/Sheets or institution-approved cloud office suite; four test collaborators where available.

Underlying theory

Cloud collaboration uses permissions, comments, suggestions and revision history. Version history records edit times and editors; restoration creates a recoverable current state rather than erasing the history.

Safety, privacy and quality

Share with the minimum required permission. Remove public-link access after the activity and never share confidential documents through unrestricted links.

Procedure

1. Create or open a shared lab-report sheet.
2. Assign Viewer, Commenter or Editor access deliberately.
3. Make identifiable edits from different accounts.
4. Delete a sample table or value such as B5.
5. Open File > Version history > See version history.
6. Use time stamps and color-coded editor identities to locate the change.
7. Restore the correct version and verify the table.
8. Optional: create a one-page Overleaf document with title, formula and list.

Observation / evidence

Record sharing level, editor colors, selected version time, restored content and final verification.

Result

The team identifies an editor's change and restores deleted content using version history.

Possible errors and troubleshooting

- Sharing "Anyone with the link can edit".
- Restoring without first checking what later valid edits will be affected.
- Using comments as a substitute for required data.

Viva questions with answers

Where is version history in Google Sheets?

File > Version history > See version history.

How do colors help?

They associate edits with specific collaborators in the selected version.

What is least privilege?

Granting only the access level required for the task.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Series Test 1

Official row: CO2 · Apply · 2 hours. The page-4 assessment table describes CA1 as the practical test for the first half of experiments. The exact task selection is not specified; follow faculty instructions.

Expected / Practice Questions

Very Short Answer

What does COUNT count?

Answer / model points

Numeric cells. Use COUNTA for all non-empty cells, including text names.

Formula

Write the formula to mark B2 as Selected when runs are at least 50.

Answer / model points

=IF(B2>=50,"Selected","Waitlist")

Short Answer

Why is Slide Master used?

Answer / model points

It controls repeated layout, branding, name/roll number and placeholders consistently across slides.

Application

Explain how to restore a deleted table in Google Sheets.

Answer / model points

Open File > Version history > See version history, select the correct timestamp/editor state, inspect the preview, restore the version and verify the recovered table.

Module II summary

Convert raw information into structured documents, correct spreadsheets, clear presentations and recoverable collaborative work. Complete every experiment with evidence, result, troubleshooting notes and faculty verification; memorising menu paths alone is insufficient.

Data Analysis, Visualization and AI Presentations

Turn clean data into summaries, filters, charts and verified professional presentations.

Official module title: Not specified in the supplied syllabus PDF; this is a descriptive handbook navigation title.

6 instructional hours

CO3

Apply

Malayalam support: PivotTable source data delete ചെയ്യാതെ summary ഉണ്ടാക്കുന്നു. Slicer visual filter ആണ്. Chart തിരഞ്ഞെടുക്കുന്നത് data relationship അനുസരിച്ചായിരിക്കണം; decoration അനുസരിച്ച് അല്ല.

Concept handbook

Preparing data for analysis

Remove merged cells, fix headers, standardize categories and make numbers numeric. Document missing values rather than hiding them.

PivotTable fields

Rows and Columns group categories; Values aggregate; Filters and slicers restrict the view. Always check whether the aggregation is Sum, Count, Average or another function.

Slicer behavior

A slicer changes the visible subset and therefore totals. It does not delete source records. Clear the filter to return to the full dataset.

Refresh and source range

When new rows are added, refresh. If the source is not a dynamic table, extend the source range before refreshing.

Chart selection

Bar/column: category comparison. Line: ordered time trend. Pie: small part-to-whole composition. Scatter: relationship between two numeric variables.

Honest visualization

Use clear titles, units and readable labels. Avoid misleading axes, unnecessary 3D effects, excessive colors and hidden categories.

AI slide workflow

Specify audience, objective, structure, slide count, tone and evidence constraints. Generate, fact-check, edit, cite and present.

AI-assisted current information

Prices and current specifications change. Record retrieval date and verify using primary manufacturer or retailer sources before making recommendations.

Interactive Pivot/Slicer demonstration

Select a region. The sample category totals and Grand Total update without deleting source data.

Grand Total: 976



Practical / Activity / Experiment Centre

Experiment 8

Data Analysis

2 instructional hours · CO3 · Apply

Official practical outcome: Use PivotTables and slicers to analyze a given spreadsheet dataset.

Aim

Summarize a 50-row sales dataset by category and interactively filter it by region.

Apparatus / software

Spreadsheet application with PivotTable and slicer support; clean sales dataset.

Underlying theory

A PivotTable groups records and aggregates numeric fields. Rows define categories, Values define summaries and slicers provide a visible filter. New source rows require a refresh and sometimes a source-range update.

Safety, privacy and quality

Use a copy of the dataset. Check for personal data before sharing or uploading.

Procedure

- 1. Inspect headers and remove blank rows/merged cells.
- 2. Convert the range to a table where supported.
- 3. Insert a PivotTable.
- 4. Drag Category to Rows and Sales to Values.
- 5. Verify that Values uses Sum, not Count.

6. Insert a Region slicer and select South.
7. Record the filtered Grand Total.
8. Add a new source row and use Refresh to update the PivotTable.

Observation / evidence

Record category totals, unfiltered Grand Total, South-filtered total and refreshed total.

Result

A PivotTable with a Region slicer is created and refreshed correctly.

Possible errors and troubleshooting

- Using text-formatted numbers.
- Including blank header cells.
- Forgetting Refresh after changing source data.

Viva questions with answers

What does a slicer do?

It provides an interactive visual filter for the PivotTable.

Why might Values show Count instead of Sum?

The source field may contain text, blanks or mixed data types.

How is a PivotTable updated?

Use Refresh; also ensure the source range includes new records.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Official practical outcome: Use different chart types to analyze a spreadsheet dataset.

Aim

Create pass/fail and student-marks charts and explain why each chart type is suitable.

Apparatus / software

Spreadsheet application and Foundational IT Skills series-test data.

Underlying theory

Charts encode data visually. Bar/column charts compare categories; line charts show ordered trends; pie charts show a small part-to-whole composition. Three-dimensional effects can distort perceived values and should be used cautiously.

Safety, privacy and quality

Anonymize student data before public display. Do not manipulate axes or categories to mislead.

Procedure

1. Create a status column using `=IF(score>=40,"Passed","Failed")`.
2. Summarize counts with `COUNTIF`.
3. Insert the prescribed 3D pie chart for pass/fail percentages and note its limitations.
4. Insert a clustered horizontal bar chart with names on Y-axis and marks on X-axis.
5. Apply a chart style, including the requested dark theme.
6. Add title, labels and accessible alternative text.
7. Check that the chart matches the source values.

Observation / evidence

Record pass/fail counts and percentages, chart titles, axis assignments and verification results.

Result

Charts are created, formatted and validated against the summary table.

Possible errors and troubleshooting

- Using a pie chart for many categories.
- Removing axis labels and units.
- Starting a value axis above zero without disclosure.

Viva questions with answers

Why is a bar chart suitable for student marks?

It supports accurate comparison across named categories.

What is a misleading axis?

An axis scale that exaggerates or hides differences.

Why verify a chart against the table?

Formatting can hide data errors; the chart must represent the source correctly.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Experiment 10

AI Slide Generation

2 instructional hours · CO3 · Apply

Official practical outcome: Use Generative AI platforms to transform structured text prompts into professional multi-slide presentations.

Aim

Generate and critically review an AI-assisted presentation from a structured prompt.

Apparatus / software

Gamma.app or institution-approved equivalent; Manus AI or approved research agent where available; browser and stopwatch.

Underlying theory

Generative AI predicts content from prompts and context. It may hallucinate facts, prices, sources and images. A professional workflow separates prompt design, generation, verification, editing and citation.

Safety, privacy and quality

Never upload private student data, company drawings, passwords, proprietary specifications or copyrighted content without permission. Verify external claims using primary sources.

Procedure

1. Write a structured prompt specifying audience, goal, eight slides, tone, sections and source requirements.
2. Generate the deck and record elapsed time in seconds.
3. Check each claim, image and citation; edit unsupported content.
4. For the laptop-price activity, request a table for Dell, HP and Lenovo models with 16 GB RAM.
5. Record the retrieval date and verify prices on primary retailer/manufacturer sources.
6. Write a five-sentence recommendation that states assumptions and budget limits.
7. Create a five-slide presentation from the verified table.

Observation / evidence

Record tool, prompt, generation time, claims corrected, source checks and final slide count.

Result

An AI-generated presentation is reviewed, corrected and documented.

Possible errors and troubleshooting

- Treating AI output as verified research.
- Requesting “make a presentation” without audience or constraints.
- Using current prices without retrieval date and source verification.

Viva questions with answers

What is hallucination?

A plausible-looking but unsupported or false AI output.

What makes a prompt structured?

It specifies context, role, task, output format and constraints.

Why record generation time?

It allows comparison of workflow efficiency and documents the experiment.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Expected / Practice Questions

Very Short Answer

What button updates a PivotTable after source data changes?

Answer / model points

Refresh.

Short Answer

Why can a numeric PivotTable field show Count rather than Sum?

Answer / model points

The source field may contain text, blanks or mixed data types, so the spreadsheet treats it as non-numeric.

Comparison

Choose chart types for marks by student and monthly attendance trend.

Answer / model points

Horizontal bar/column for marks by named student; line chart for attendance over ordered months.

Higher Order

What must be verified in an AI-generated current laptop-price table?

Answer / model points

Exact model, 16 GB RAM condition, seller/manufacturer, price, currency, tax/shipping context, stock status and retrieval date.

Module III summary

Turn clean data into summaries, filters, charts and verified professional presentations. Complete every experiment with evidence, result, troubleshooting notes and faculty verification; memorising menu paths alone is insufficient.

GPT-Guided Learning and Graphic Design

Use generative AI as a controlled learning and design assistant, not as an unquestioned authority.

Official module title: Not specified in the supplied syllabus PDF; this is a descriptive handbook navigation title.

6 instructional hours

CO4

Apply

Malayalam support: Generative AI output draft മാത്രമാണ്. Prompt clear ആക്കണം, technical facts verify ചെയ്യണം, personal/company confidential data public AI tool-ലേക്ക് upload ചെയ്യരുത്.

Concept handbook

How GPT-guided learning should work

The assistant explains at the learner's level, asks a checkpoint question, waits, gives feedback, adapts difficulty and states uncertainty.

C-R-T-O prompt framework

Context: situation and learner. Role: perspective. Task: exact action. Output: format, depth, constraints and stopping rule.

One-question-at-a-time interview

Sequential questioning prevents answer leakage and supports scoring. A rubric should assess correctness, clarity and application.

Verification and academic integrity

AI can be wrong. Check against textbooks, official documentation or faculty guidance. Do not submit unedited generated text as personal work.

Privacy and engineering confidentiality

Never upload passwords, personal identifiers, examination data, company drawings, unpublished designs or client information to public AI tools.

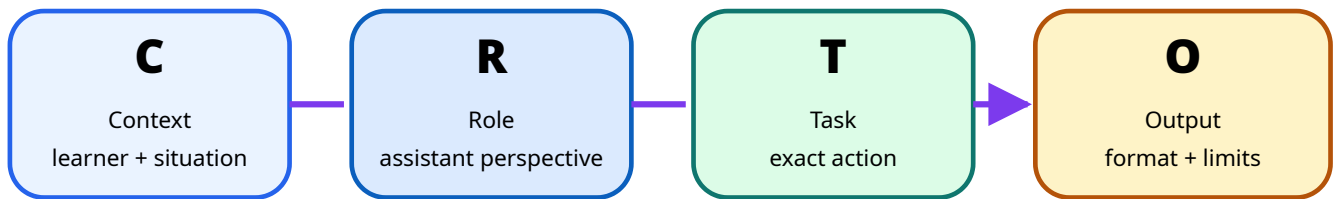
Graphic design fundamentals

Use hierarchy, contrast, alignment, repetition, proximity and whitespace. Keep font count low and ensure text remains readable on mobile and print.

AI-assisted image review

Check consent, copyright, realism, text artifacts, component geometry, cable routing, perspective and safety details before publishing.

C-R-T-O prompt construction



Example: Context: Semester-I learner revising RAM. Role: patient IT lab tutor. Task: explain RAM versus SSD, then ask one question. Output: 120 words, one comparison table, wait for my answer before continuing.

Practical / Activity / Experiment Centre

Experiment 11

Generative Pre-trained Transformer (GPT) Guided Learning

2 instructional hours · CO4 · Apply

Official practical outcome: Create a “Study Assistant” prompt for technical concepts.

Aim

Create and test a guided-learning prompt and a one-question-at-a-time technical interview.

Apparatus / software

ChatGPT, Gemini or institution-approved generative AI tool; prompt record sheet.

Underlying theory

A GPT generates responses from patterns in training and provided context. The C-R-T-O framework defines Context, Role, Task and Output. Good tutoring prompts request staged explanation, checks for understanding, feedback and uncertainty disclosure.

Safety, privacy and quality

Do not enter confidential engineering data or personal information. Verify technical claims and calculations independently.

Procedure

1. Give the interviewer prompt: Act as a technical interviewer for a Junior Engineer role. Ask five computer questions one by one.
2. Answer each question before requesting the next.
3. Ask for a rubric-based score and corrected model answer.
4. Define C-R-T-O and write a new use-case prompt.

5. Create a Study Assistant prompt that teaches one concept, asks a checkpoint question and adapts to the learner.
6. Test, revise and record one weak and one improved prompt.

Observation / evidence

Record prompt versions, model outputs, corrections, verification source and learning reflection.

Result

A controlled Study Assistant and technical interview prompt are created and evaluated.

Possible errors and troubleshooting

- Combining unrelated tasks in one vague prompt.
- Accepting corrections without independent verification.
- Failing to specify output format or learner level.

Viva questions with answers

What does C-R-T-O stand for?

Context, Role, Task and Output.

Why ask one question at a time?

It supports interaction, feedback and controlled difficulty.

Can a GPT guarantee correctness?

No. Outputs require verification, especially for technical or current facts.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Official practical outcome: Design a technical event poster using AI tools.

Aim

Create a college-event logo and a professional technical poster using a web-based graphic-design platform.

Apparatus / software

Canva or approved equivalent, institution-approved images, browser.

Underlying theory

Design quality depends on hierarchy, alignment, contrast, repetition, proximity and whitespace. AI background or image tools require technical and visual review; generated content may contain unrealistic equipment or text artifacts.

Safety, privacy and quality

Use consented photographs. Do not create deceptive composites or use copyrighted logos/assets without permission. Review generated backgrounds for unsafe or technically impossible details.

Procedure

1. Define event name, audience, size and required details.
2. Create a simple art-festival logo that remains readable at small size.
3. Build a technical event poster with clear title, date, venue, organizer and call to action.
4. Use Magic Studio or equivalent to place a consented portrait before a high-tech server-room background.
5. Inspect edges, lighting, equipment realism and text accuracy.
6. Export screen and print versions using suitable dimensions and file formats.

Observation / evidence

Record canvas size, hierarchy, font count, contrast check, source/consent status and export settings.

Result

A technically reasonable, readable and ethically produced event design is exported.

Possible errors and troubleshooting

- Using too many fonts.
- Placing critical text over a busy image.
- Accepting AI-generated servers, cables or safety details without review.

Viva questions with answers

What is visual hierarchy?

A design order that guides attention from most to least important information.

Why verify AI-generated backgrounds?

They can contain impossible equipment, distorted objects or unsafe details.

What is whitespace?

Intentional empty space that improves grouping and readability.

Record-writing format

Date: _____ System/File: _____ Faculty verification: _____

Aim: _____

Procedure summary: _____

Observation / screenshot reference: _____

Result and reflection: _____

Series Test 2

Official row: CO4 · Apply · 2 hours. The page-4 assessment table describes CA2 as the practical test for the second half of experiments. The exact task selection is not specified; follow faculty instructions.

Expected / Practice Questions

Very Short Answer

Expand C-R-T-O.

Answer / model points

Context, Role, Task, Output.

Prompt Design

Write one output constraint for a Study Assistant prompt.

Answer / model points

Ask one question at a time, wait for my response, then give feedback in a three-column table: concept, my answer, correction.

Ethics

Why should a confidential company drawing not be uploaded to a public AI tool?

Answer / model points

It can breach company policy, intellectual-property obligations and client confidentiality, and the provider may process or retain the data.

Design

State four poster-design principles.

Answer / model points

Hierarchy, contrast, alignment and whitespace; repetition and proximity are also relevant.

Module IV summary

Use generative AI as a controlled learning and design assistant, not as an unquestioned authority. Complete every experiment with evidence, result, troubleshooting notes and faculty verification; memorising menu paths alone is insufficient.

OPEN-ENDED EXPERIMENT

Student-designed application beyond the prescribed list

The official objective is to encourage creativity, innovation and application of knowledge. Each student or group conceives, designs and implements an experiment related to the course outcomes. It may extend an existing experiment, integrate multiple concepts, build a mini-setup/prototype or compare the performance of a system/component.

Planning

1. Define a real problem and user.
2. Map the idea to one or more COs.
3. State deliverables, data and success criteria.
4. Identify privacy, copyright and safety risks.

Possible course-aligned ideas

- Automated lab-record status dashboard.
- PC inventory and port-identification guide.
- Version-history recovery comparison across two cloud tools.
- Verified AI-assisted technical-event communication package.

Required record

- Aim, scope and plan.
- Tools/data and permissions.
- Procedure and screenshots.
- Observations and analysis.
- Result, limitation and team contribution.

Do not invent operational limits: exact account permissions, organization data and institution-owned assets must be supplied or approved by the faculty. Public AI tools must not receive confidential college or company information.

Official Open-ended Experiment evaluation - 50 marks		
Category	What is evaluated	Marks
Originality and relevance	Creativity, innovation and relevance to course objectives.	10
Clarity of objectives and plan	Clear aim, scope and systematic experimental plan.	10
Execution and data recording	Accuracy, safety and neatness.	10
Analysis, interpretation and presentation	Logical analysis and presentation of results.	10
Teamwork and innovation demonstrated	Collaboration, initiative and creativity.	10

FORMULA, RULE AND PROCEDURE BANK

Commands and decisions worth revising

Spreadsheet formulas

Cell references

=SUM(B2:B21)

=AVERAGE(B2:B21)

=COUNTIF(C2:C21,"Batsman")

=IF(B2>=50,"Selected","Waitlist")

- A1 - relative row and column.
- \$A\$1 - absolute row and column.
- \$A1 - fixed column.
- A\$1 - fixed row.

Warning: COUNT counts numbers; COUNTA counts non-empty names.

Email fields

- **To:** primary action recipient.
- **CC:** visible information copy.
- **BCC:** hidden recipient list.

Chart choice

- Bar/column - comparison.
- Line - trend over ordered time.
- Pie - few parts of one whole.
- Scatter - two-number relationship.

PivotTable procedure

1. Clean source data.
2. Insert PivotTable.
3. Place Category in Rows.
4. Place Sales in Values and verify Sum.
5. Add Region slicer.
6. Refresh after source changes.

C-R-T-O

- Context - learner and situation.
- Role - assistant perspective.
- Task - exact action.
- Output - structure, limit and stopping rule.

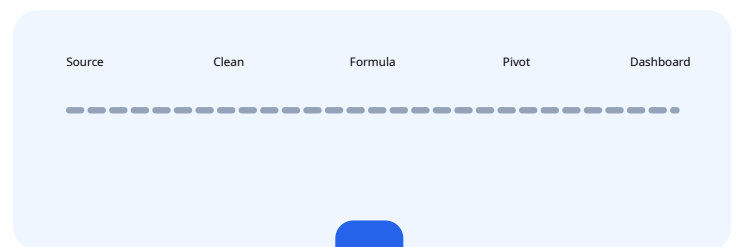
DIAGRAM AND VISUAL LIBRARY

One-look process diagrams

Professional file lifecycle



Data-to-dashboard animation



Static print state: data moves from a clean source through analysis to a verified dashboard.

SOLVED EXAMPLES AND DEMONSTRATIONS

Complete working, not unexplained answers

1. Cricket selection formula

1. **Given:** Runs in B2 = 62; cut-off = 50.
2. **Required:** Selection Status.
3. **Formula:** =IF(B2>=50,"Selected","Waitlist")
4. **Test:** 62 >= 50 is TRUE.
5. **Final output:** Selected.
6. **Verification:** Change B2 to 49; output must become Waitlist.

2. COUNT versus COUNTA

1. **Given:** A2:A21 contains 20 player names.
2. **Incorrect assumption:** =COUNT(A2:A21) gives 20.
3. **Reason:** COUNT ignores text.
4. **Correct formula:** =COUNTA(A2:A21)
5. **Final output:** 20, provided no blanks exist.

3. Pass/fail summary

1. Status formula: =IF(B2>=40,"Passed","Failed").
2. Passed count: =COUNTIF(C2:C31,"Passed").
3. Failed count: =COUNTIF(C2:C31,"Failed").
4. Check: Passed + Failed must equal the number of valid scores.
5. Use the summary table, not raw rows, as the pie-chart source.

4. Version-history recovery

1. Open File > Version history > See version history.
2. Select the time before deletion.
3. Use editor colors and the preview to verify the correct state.
4. Restore only after considering valid later edits.
5. Confirm the table and document the recovery.

5. PivotTable verification

1. Manually total a small sample category.
2. Create the PivotTable using Category in Rows and Sales in Values.
3. Confirm the aggregation label is Sum of Sales.
4. Compare the PivotTable category total with the manual sample.
5. Filter South using the slicer and record the changed Grand Total.

6. C-R-T-O Study Assistant prompt

1. **Context:** Semester-I student learning storage.
2. **Role:** IT lab tutor.
3. **Task:** Compare RAM, HDD and SSD, then ask one question.
4. **Output:** 150 words, one table, no next topic until the learner answers.
5. **Verification:** Check technical claims against the official text/reference.

STUDENT ACTIVITIES FOR SELF-LEARNING

Official 15-hour activity plan

Self-learning activities			
Module	Experiment	Official activity coverage	Hours
I	1	Draw or photograph the ports of the lab PC and label them; compare RAM, processor and SSD/HDD configuration of another smart device with the lab PC; list five system and five application software items.	1
I	2	Create a semester directory tree using subject codes and module subfolders; perform the complete Backpack/Food/Tools file-management scenario.	1
I	3	Use filetype:pdf and site:edu search operators; enroll in the course LMS; draft and send the formal schedule-change email with correct netiquette and private recipient handling.	1
II	4	Create an Assignment Cover Sheet using Heading styles and a professional internship biodata with Normal margins, borderless two-column personal-details table and photo placeholder; export to PDF.	1
II	5	Apply COUNTIF for pass count; build the cricket dataset calculations using SUM, AVERAGE, COUNTIF, IF and the prescribed participant-count check.	1
II	6	Edit Slide Master so name and roll number appear automatically on every slide.	1
II	7	Explain and demonstrate version-history recovery and editor identification in Google Sheets; optionally create a one-page Overleaf document with title, mathematical formula and list.	2
III	8	Create a PivotTable from a 50-row sales dataset, add a Region slicer, observe the South-filtered Grand Total and refresh after adding a source row.	2
III	9	Create pass/fail summary and 3D pie chart; create a clustered horizontal bar chart for student marks and apply a dark chart style.	2
III	10	Generate an eight-slide deck and record generation time; research three 16 GB RAM laptop brands, verify a price table, write a five-sentence recommendation and make a presentation.	1
IV	11	Run a five-question one-by-one Junior Engineer computer interview; explain C-R-T-O and create a C-R-T-O prompt for a chosen use case.	1
IV	12	Design a college art-festival logo and create a professional poster using a consented portrait and a high-tech server-room background.	1
Total self-learning hours			15

ASSESSMENT AND RECORD QUALITY

Official scheme preserved without inventing a corrected split

Assessment Methodology - Lab, as printed						
Row	Attendance	CE - Continuous evaluation	CA1	CA2	OEE	ESE
Mode	Attendance	Continuous evaluation	Practical test - first half	Practical test - second half	Open-ended experiment	Practical
Duration	-	-	2 hours	2 hours	-	3 hours
Exam mark	-	50	40	40	50	40
Converted to	-	-	15	15	10	-
Marks row	5	30	-	-	10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	Not printed

Practical Continuous Internal Evaluation - 50 marks	
Criterion	Marks
Preparation & punctuality	10
Experimental setup & procedure	10
Observation & data recording	5
Analysis & result interpretation	10
Viva voce / concept understanding	10
Workmanship, discipline & teamwork	5

Practical Test and ESE - 40 marks	
Category	Marks
Write-up / procedure before experiment	10
Experiment setup & execution	10
Observation & result / output	8
Viva voce	8
Record	4

EXPECTED / PRACTICE QUESTIONS

Module-wise answer bank

These are handbook practice questions, not official SITTR questions. The official syllabus supplies experiments and evaluation schemes, not a model question paper.

Module I

Very Short Answer

Differentiate RAM and storage.

Answer / model points

RAM is volatile working memory used by active programs; storage such as SSD/HDD is non-volatile and preserves files after power-off.

Short Answer

Why should student email addresses be placed in BCC in the official scenario?

Answer / model points

BCC prevents recipients from seeing one another's private email addresses while still delivering the same announcement.

Application

Write a search query for PDF resources about Computer Engineering on educational domains.

Answer / model points

filetype:pdf "Computer Engineering" site:edu

Troubleshooting

A copied file cannot be found. What should be checked?

Answer / model points

Check the destination path, current folder, hidden-file setting, filename/extension and whether Move was accidentally used instead of Copy.

Module II

Very Short Answer

What does COUNT count?

Answer / model points

Numeric cells. Use COUNTA for all non-empty cells, including text names.

Formula

Write the formula to mark B2 as Selected when runs are at least 50.

Answer / model points

=IF(B2>=50,"Selected","Waitlist")

Short Answer

Why is Slide Master used?

Answer / model points

It controls repeated layout, branding, name/roll number and placeholders consistently across slides.

Application

Explain how to restore a deleted table in Google Sheets.

Answer / model points

Open File > Version history > See version history, select the correct timestamp/editor state, inspect the preview, restore the version and verify the recovered table.

Module III

Very Short Answer

What button updates a PivotTable after source data changes?

Answer / model points

Refresh.

Short Answer

Why can a numeric PivotTable field show Count rather than Sum?

Answer / model points

The source field may contain text, blanks or mixed data types, so the spreadsheet treats it as non-numeric.

Comparison

Choose chart types for marks by student and monthly attendance trend.

Answer / model points

Horizontal bar/column for marks by named student; line chart for attendance over ordered months.

Higher Order

What must be verified in an AI-generated current laptop-price table?

Answer / model points

Exact model, 16 GB RAM condition, seller/manufacturer, price, currency, tax/shipping context, stock status and retrieval date.

Module IV

Very Short Answer

Expand C-R-T-O.

Answer / model points

Context, Role, Task, Output.

Prompt Design

Write one output constraint for a Study Assistant prompt.

Answer / model points

Ask one question at a time, wait for my response, then give feedback in a three-column table: concept, my answer, correction.

Ethics

Why should a confidential company drawing not be uploaded to a public AI tool?

Answer / model points

It can breach company policy, intellectual-property obligations and client confidentiality, and the provider may process or retain the data.

Design

State four poster-design principles.

Answer / model points

PRACTICE MODEL PAPER - NOT AN OFFICIAL SITTTR MODEL QUESTION PAPER

Practical preparation set

Part A - Very Short Answer

1. Differentiate system software and application software.
2. Write one advanced search operator.
3. What does \$A\$1 mean?
4. State one purpose of a slicer.
5. Expand C-R-T-O.

Part B - Short / Procedure

1. Write the steps to restore an accidentally deleted Google Sheets table.
2. Explain COUNT, COUNTA and COUNTIF with examples.
3. Give a chart-selection rule for category comparison and time trend.
4. Write an ethical Study Assistant prompt.

Part C - Practical Tasks

1. Create the prescribed cricket spreadsheet and verify all formulas.
2. Create a PivotTable with Region slicer and record Grand Totals.
3. Create a five-slide pitch deck using Slide Master.
4. Prepare a technical event poster and document consent, source and export settings.

Assessment approach: use the official practical-test/ESE 40-mark criteria: write-up 10, setup/execution 10, observation/result 8, viva 8 and record 4.

QUICK REVISION

Final checklist before practical tests

One-look rules

- CPU executes; RAM works; SSD/HDD stores.
- Copy duplicates; Move relocates.
- To acts; CC informs; BCC hides addresses.
- Styles create consistent documents and TOCs.
- COUNT numbers; COUNTA non-empty cells.
- Refresh PivotTables after source changes.
- Choose charts by relationship, not decoration.
- AI output requires verification and privacy review.

Common failure points

- Wrong file/folder destination.
- Private addresses exposed in CC.

- Text numbers causing PivotTable Count.
- Formula ranges include headers or miss rows.
- Version history restored without reviewing later edits.
- Misleading chart axes or unnecessary 3D.
- AI prices used without date/source.
- Poster uses unlicensed image or unreadable contrast.

GLOSSARY

Technical terms

CPU

Processor that executes instructions.

RAM

Volatile working memory.

SSD

Flash-based non-volatile storage.

GUI

Graphical User Interface.

Netiquette

Professional and safe online communication behaviour.

Cell reference

Spreadsheet address such as B2 or \$B\$2.

PivotTable

Interactive grouped summary of source records.

Slicer

Visible filter control for a table/PivotTable.

Version history

Record of file changes, times and editors.

GPT

Generative Pre-trained Transformer.

Hallucination

Unsupported or false AI output presented plausibly.

Visual hierarchy

Ordering design elements by importance.

EQUIPMENT, SOFTWARE AND REFERENCES

Official learning resources

Major equipment/components for a batch of 30		
Particular	Specification	Quantity
PC	Intel Core i5/AMD Ryzen 5, 12th Gen or higher; 16 GB RAM; 512 GB SSD; 18.5-inch LED monitor.	30
Internet	Dedicated leased line or broadband, minimum 100 Mbps, Wi-Fi 6 support.	1
Networking Kit	24-port managed switch, CAT6 patch cords and RJ45 connectors.	2
Hardware Demo Kit	Disassembled motherboard, RAM, HDD, SSD and SMPS components.	2
External Peripherals	HDMI, VGA, DisplayPort, USB-C cables and 1080p webcams.	2

Major software for a batch of 30		
Particular	Specification	Quantity
Operating System	Windows 10/11 Professional or latest Ubuntu LTS.	30
Office Productivity	Microsoft Office 365 Academic or LibreOffice/OpenOffice.	30
Web Browsers	Latest Chrome, Firefox and Edge.	30 installs
Cloud Platforms	Access to Google Workspace, Overleaf (LaTeX).	“Cloud Platforms” (as printed in the quantity cell)
AI & Design Tools	Access to ChatGPT/Gemini and AI-powered Design platforms.	Not specified in the supplied official syllabus PDF.

Official textbook and reference

- *Computer Fundamentals* - P. K. Sinha, BPB Publications.
- *IT Essentials: PC Hardware and Software* - Cisco Networking Academy, Cisco Press.

Official web resources

- Overleaf Learn - Module II.
- Gamma - Module III.
- Manus AI - Module III.
- Canva - Module IV.

These links are curriculum references, not runtime dependencies of this offline-capable page.